Introduction to conjoint analysis

Welcome!

Housekeeping!

- Be kind and curious!
- Slack and Zoom chat
- Ask questions

Schedule

Day 1

Time	Activity
10:30–12:30	Welcome + Intro to conjoint analysis
12:30–13:30	*Break*
13:30–15:00	Designing conjoint experiments

Day 2

Time	Activity
10:30–12:30	Conjoint data and causal inference (part 1)
12:30–13:30	*Break*
13:30–15:00	Conjoint data and causal inference (part 2)

Schedule

Day 3

Time	Activity
10:30–12:30	Conjoint data and respondent preferences (part 1)
12:30–13:30	*Break*
13:30–15:00	Conjoint data and respondent preferences (part 2)

Day 4

Time	Activity
10:30–12:30	Implementing and fielding conjoint surveys (part 1)
12:30–13:30	*Break*
13:30–15:00	Implementing and fielding conjoint surveys (part 2)

About me

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- Assistant professor of ~~public policy~~ international relations, ~~Georgia State University~~ Georgetown University in Qatar
- Data visualization, statistics, and causal inference

Meeting you where you are

This course is designed for someone who:

- Is familiar with R (and ideally tidyverse)
- Has some experience with statistical modeling with linear models
- Maybe has an idea for a conjoint experiment

You'll learn:

- What conjoint analysis is
- How to measure causal effects and explore "product" and "consumer" preferences
- How to design and run your own conjoint experiments

Course structure

My turn

- Lecture segments
- Feel free to just watch, take notes, browse docs, or tinker around with the code
- Lots of live-coding! Most content lives in Quarto files

Your turn

- Exercises for you to do
- Work on your own or with others

Bring your own ideas!

- The best way to learn is by building something
- I’ve given you playground code, but it’s fairly basic. *This is by design!*
- You’ll have time during the “Your turn”s to to play around and experiment. Try to use these new principles and skills to create something!

Getting help

Use Zoom reactions

- ✖ =
  "I'm stuck and need help!"
- ✔ =
  "I finished the exercise"

Ask longer, more detailed questions in Slack

Your turn

Introduce yourself:

- Name and professional affiliation
- On a scale of 1–10, how well do you know…
  - R?
  - Tidyverse?
  - Statistical models like OLS, multinomial logit, and Bayesian methods?
  - Conjoint analysis?
- What do you hope to get out of this course?

04:00

What is conjoint analysis?

The problem of preferences

How do you measure what people *really* want?

- **Direct questions:** "Do you care about the environment?" → Everyone says yes
- **Stated vs. revealed preference:** What people say ≠ what they do
- **Single attributes:** "Is healthcare important?" Of course! But *how* important vs. cost? Do they interact?
- **Ranking:** People rank things differently depending on context

People make trade-offs. Survey experiments that don't force trade-offs can give misleading answers.

Ice cream!

It’s hot outside and you want to buy some ice cream. You go to your local ice cream shop and find a menu with all these possibilities:

- Vanilla, chocolate, or strawberry
- Served in a bowl or served in a waffle cone
- No extra toppings, topped with nuts, or topped with sprinkles
- Priced at $3, $4, or $5

Which kind do you want? Why?

No ice cream maximizes everyone’s happiness.

There are too many trade-offs

What is conjoint analysis?

> Conjoint analysis is a family of survey-based techniques where respondents evaluate or choose among hypothetical profiles described by multiple attributes, revealing their preferences through the trade-offs they make.

conjoint = *considered jointly*

Respondents evaluate attributes **together** (as bundles), not separately

Use statistics to uncover preferences from these bundles andhs.co/icecream

Where did it come from?

Psychology (1960s)

Luce, R. Duncan, and John W. Tukey. 1964. "Simultaneous Conjoint Measurement: A New Type of Fundamental Measurement." *Journal of Mathematical Psychology* 1 (1): 1–27. https://doi.org/10.1016/0022-2496(64)90015-X.

Originally developed to measure preferences of sets of features

JOURNAL OF MATHEMATICAL PSYCHOLOGY: 1, 1-27 (1964)

Simultaneous Conjoint Measurement:
A New Type of Fundamental Measurement

R. DUNCAN LUCE[1, 2]

University of Pennsylvania, Philadelphia, Pennsylvania

AND

JOHN W. TUKEY[1, 3]

Princeton University, Princeton, New Jersey, and Bell Telephone Laboratories, Murray Hill, New Jersey

The essential character of what is classically considered, e.g., by N. R. Campbell, the fundamental measurement of extensive quantities is described by an axiomatization for the comparision of effects of (or responses to) arbitrary *combinations* of "quantities" of a *single specified kind.* For example, the effect of placing one arbitrary combination of masses on a pan of a beam balance is compared with another arbitrary combination on the other pan. Measurement on a ratio scale follows from such axioms. In this paper, the essential character of simultaneous *conjoint* measurement is described by an axiomatization for the comparision of effects of (or responses to) *pairs* formed from *two specified kinds* of "quantities." The axioms apply when, for example, the effect of a pair consisting of one mass and one difference in gravitational potential on a device that responds to momentum is compared with the effect of another such pair. Measurement on interval scales which have a common unit follows from these axioms; usually these scales can be converted in a natural way into ratio scales.

A close relation exists between conjoint measurement and the establishment of response measures in a two-way table, or other analysis-of-variance situations, for which the "effects of columns" and the "effects of rows" are additive. Indeed, the discovery of such measures, which are well known to have important practical advantages, may be viewed as the discovery, via conjoint measurement, of fundamental measures of the row and column variables. From this point of view it is natural to regard conjoint measurement as factorial measurement.

[1] The hospitality of the Center for Advanced Study in the Behavioral Sciences and of Stanford University is gratefully acknowledged. We wish to thank Francis W. Irwin and Richard Robinson for careful readings of an earlier version which have led to corrections and clarifications of the argument.

[2] Research supported in part by the National Science Foundation grant NSF G-17637 to the University of Pennsylvania.

[3] Research in part at Princeton University under the sponsorship of the Army Research Office (Durham).

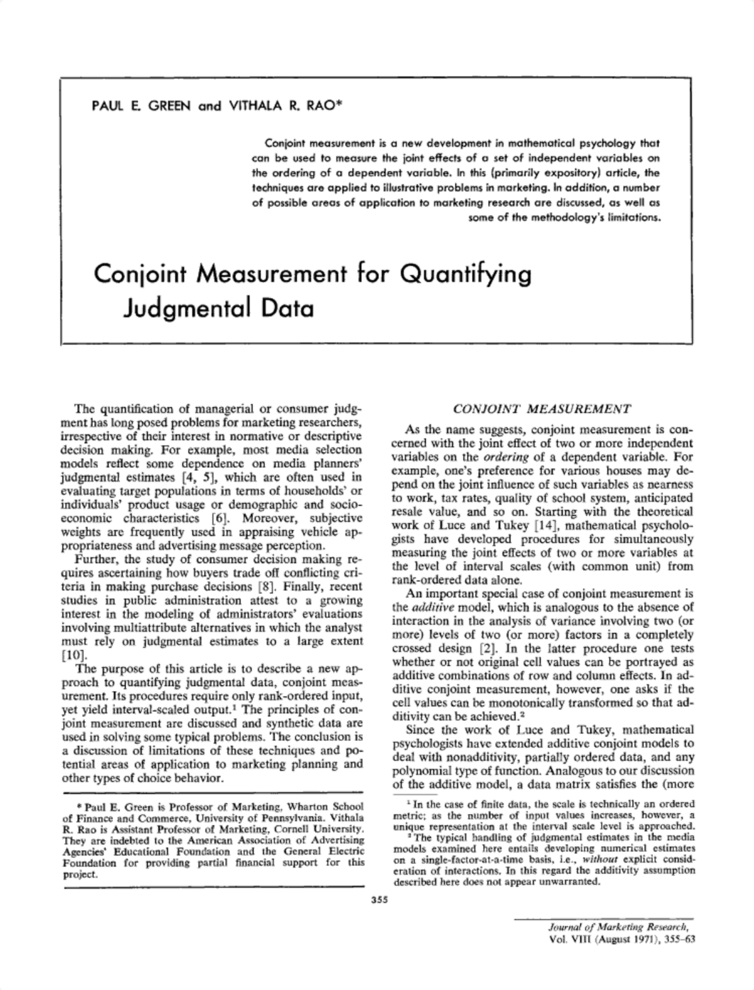
Marketing research (1970s)

Green, Paul E., and Vithala R. Rao. 1971. "Conjoint Measurement for Quantifying Judgmental Data." *Journal of Marketing Research* 8 (3): 355. https://doi.org/10.2307/3149575.

Adapted to answer marketing questions:

- How should we price our product?
- Which features should we invest in?
- What product will maximize market share?

Uses: consumer electronics, airlines, credit cards, pharmaceuticals, automobiles



PAUL E. GREEN and VITHALA R. RAO*

Conjoint measurement is a new development in mathematical psychology that can be used to measure the joint effects of a set of independent variables on the ordering of a dependent variable. In this (primarily expository) article, the techniques are applied to illustrative problems in marketing. In addition, a number of possible areas of application to marketing research are discussed, as well as some of the methodology's limitations.

Conjoint Measurement for Quantifying Judgmental Data

The quantification of managerial or consumer judgment has long posed problems for marketing researchers, irrespective of their interest in normative or descriptive decision making. For example, most media selection models reflect some dependence on media planners' judgmental estimates [4, 5], which are often used in evaluating target populations in terms of households' or individuals' product usage or demographic and socioeconomic characteristics [6]. Moreover, subjective weights are frequently used in appraising vehicle appropriateness and advertising message perception.

Further, the study of consumer decision making requires ascertaining how buyers trade off conflicting criteria in making purchase decisions [8]. Finally, recent studies in public administration attest to a growing interest in the modeling of administrators' evaluations involving multiattribute alternatives in which the analyst must rely on judgmental estimates to a large extent [10].

The purpose of this article is to describe a new approach to quantifying judgmental data, conjoint measurement. Its procedures require only rank-ordered input, yet yield interval-scaled output.[1] The principles of conjoint measurement are discussed and synthetic data are used in solving some typical problems. The conclusion is a discussion of limitations of these techniques and potential areas of application to marketing planning and other types of choice behavior.

* Paul E. Green is Professor of Marketing, Wharton School of Finance and Commerce, University of Pennsylvania. Vithala R. Rao is Assistant Professor of Marketing, Cornell University. They are indebted to the American Association of Advertising Agencies' Educational Foundation and the General Electric Foundation for providing partial financial support for this project.

CONJOINT MEASUREMENT

As the name suggests, conjoint measurement is concerned with the joint effect of two or more independent variables on the *ordering* of a dependent variable. For example, one's preference for various houses may depend on the joint influence of such variables as nearness to work, tax rates, quality of school system, anticipated resale value, and so on. Starting with the theoretical work of Luce and Tukey [14], mathematical psychologists have developed procedures for simultaneously measuring the joint effects of two or more variables at the level of interval scales (with common unit) from rank-ordered data alone.

An important special case of conjoint measurement is the *additive* model, which is analogous to the absence of interaction in the analysis of variance involving two (or more) levels of two (or more) factors in a completely crossed design [2]. In the latter procedure one tests whether or not original cell values can be portrayed as additive combinations of row and column effects. In additive conjoint measurement, however, one asks if the cell values can be monotonically transformed so that additivity can be achieved.[2]

Since the work of Luce and Tukey, mathematical psychologists have extended additive conjoint models to deal with nonadditivity, partially ordered data, and any polynomial type of function. Analogous to our discussion of the additive model, a data matrix satisfies the (more

[1] In the case of finite data, the scale is technically an ordered metric; as the number of input values increases, however, a unique representation at the interval scale level is approached.

[2] The typical handling of judgmental estimates in the media models examined here entails developing numerical estimates on a single-factor-at-a-time basis, i.e., *without* explicit consideration of interactions. In this regard the additivity assumption described here does not appear unwarranted.

Journal of Marketing Research, Vol. VIII (August 1971), 355–63

Apple vs. Samsung (2012)

- Apple sued Samsung for patent infringement: copied features like sliding to unlock, search, autocorrect, pinch-to-zoom, etc.
- Apple hired MIT marketing professor John Hauser to measure value of those features
- Conjoint experiment analyzed with a hierarchical Bayesian model
- Consumers would be willing to spend between $32–$102 extra for these features
- These results (and other bits of evidence) scaled up to $2 billion in damages awarded to Apple

Causal inference (2010s)

Hainmueller, Jens, Daniel J. Hopkins, and Teppei Yamamoto. 2014. "Causal Inference in Conjoint Analysis: Understanding Multidimensional Choices via Stated Preference Experiments." *Political Analysis* 22 (1): 1–30. https://doi.org/10.1093/pan/mpt024.

Instead of analyzing a whole constellation of attributes, look at specific levers individually.

Conjoints are like RCTs, just fancier.

Since treatment assignment is random, we can find the effect on the selection of probability (or favorability) when changing an attribute.

Uses: Measure effects of political candidate characteristics, issue salience, organization characteristics


Advance Access publication December 19, 2013
Political Analysis (2014) 22:1–30
doi:10.1093/pan/mpt024

Causal Inference in Conjoint Analysis: Understanding Multidimensional Choices via Stated Preference Experiments

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Edited by R. Michael Alvarez

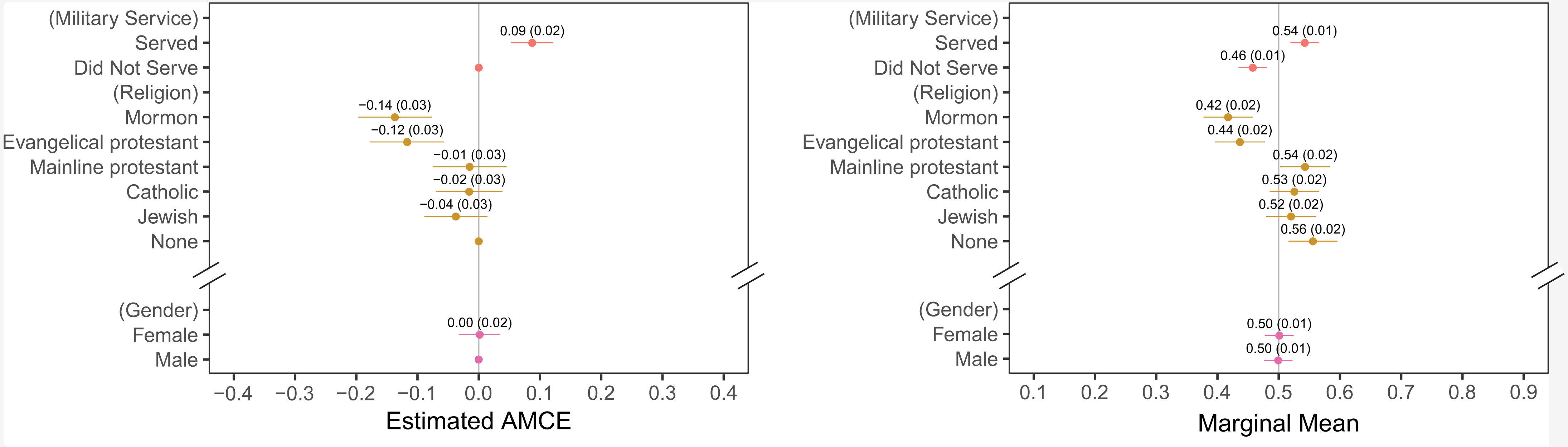
Survey experiments are a core tool for causal inference. Yet, the design of classical survey experiments prevents them from identifying which components of a multidimensional treatment are influential. Here, we show how *conjoint analysis*, an experimental design yet to be widely applied in political science, enables researchers to estimate the causal effects of multiple treatment components and assess several causal hypotheses simultaneously. In conjoint analysis, respondents score a set of alternatives, where each has randomly varied attributes. Here, we undertake a formal identification analysis to integrate conjoint analysis with the potential outcomes framework for causal inference. We propose a new causal estimand and show that it can be nonparametrically identified and easily estimated from conjoint data using a fully randomized design. The analysis enables us to propose diagnostic checks for the identification assumptions. We then demonstrate the value of these techniques through empirical applications to voter decision making and attitudes toward immigrants.

1 Introduction

The study of politics is to a significant extent the study of multidimensional choices. Voters, interest groups, government officials, and other political actors form preferences and make decisions about alternatives which differ in multiple ways. To understand such choices, political scientists are increasingly employing survey experiments (Schuman and Bobo 1988; Sniderman and Grob 1996; Gaines, Kuklinski, and Quirk 2007; Sniderman 2011). Indeed, three leading political science journals published seventy-two articles with survey experiments between 2006 and 2010.[1] The primary advantage of survey experiments is their ability to produce causal conclusions. By randomly varying certain aspects of a survey across groups of respondents, researchers can ensure that systematic

Authors' note: We gratefully acknowledge the recommendations of *Political Analysis* editors Michael Alvarez and Jonathan Katz as well as the anonymous reviewers. We further thank Justin Grimmer, Kosuke Imai, and seminar participants at MIT, Harvard University, Georgetown University, and Rochester University for their helpful comments and suggestions. We are also grateful to Anton Strezhnev for excellent research assistance. An earlier version of this article was presented at the 2012 Annual Summer Meeting of the Society for Political Methodology and the 2013 Annual Meeting of the American Political Science Association. Example scripts that illustrate the estimators and companion software to embed a conjoint analysis in Web-based survey instruments are available on the authors' websites. Replication materials are available online as Hainmueller, Hopkins, and Yamamoto (2013). Supplementary materials for this article are available on the *Political Analysis* Web site.

[1]These journals are the *American Journal of Political Science*, the *American Political Science Review*, and the *Journal of Politics*.



(Military Service)
Served
Did Not Serve
(Religion)
Mormon
Evangelical protestant
Mainline protestant
Catholic
Jewish
None
(Gender)
Female
Male
0.09 (0.02)
−0.14 (0.03)
−0.12 (0.03)
−0.01 (0.03)
−0.02 (0.03)
−0.04 (0.03)
0.00 (0.02)
−0.4 −0.3 −0.2 −0.1 0.0 0.1 0.2 0.3 0.4
Estimated AMCE
0.54 (0.01)
0.46 (0.01)
0.42 (0.02)
0.44 (0.02)
0.54 (0.02)
0.53 (0.02)
0.52 (0.02)
0.56 (0.02)
0.50 (0.01)
0.50 (0.01)
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
Marginal Mean

What can we estimate with it?

Preferences

Marketing

- How much do people value the overall constellation of features?
- What product will sell best?
- What are people willing to pay for this feature?
- Estimands: **utilities, attribute importance, willingness to pay, market shares and simulations**

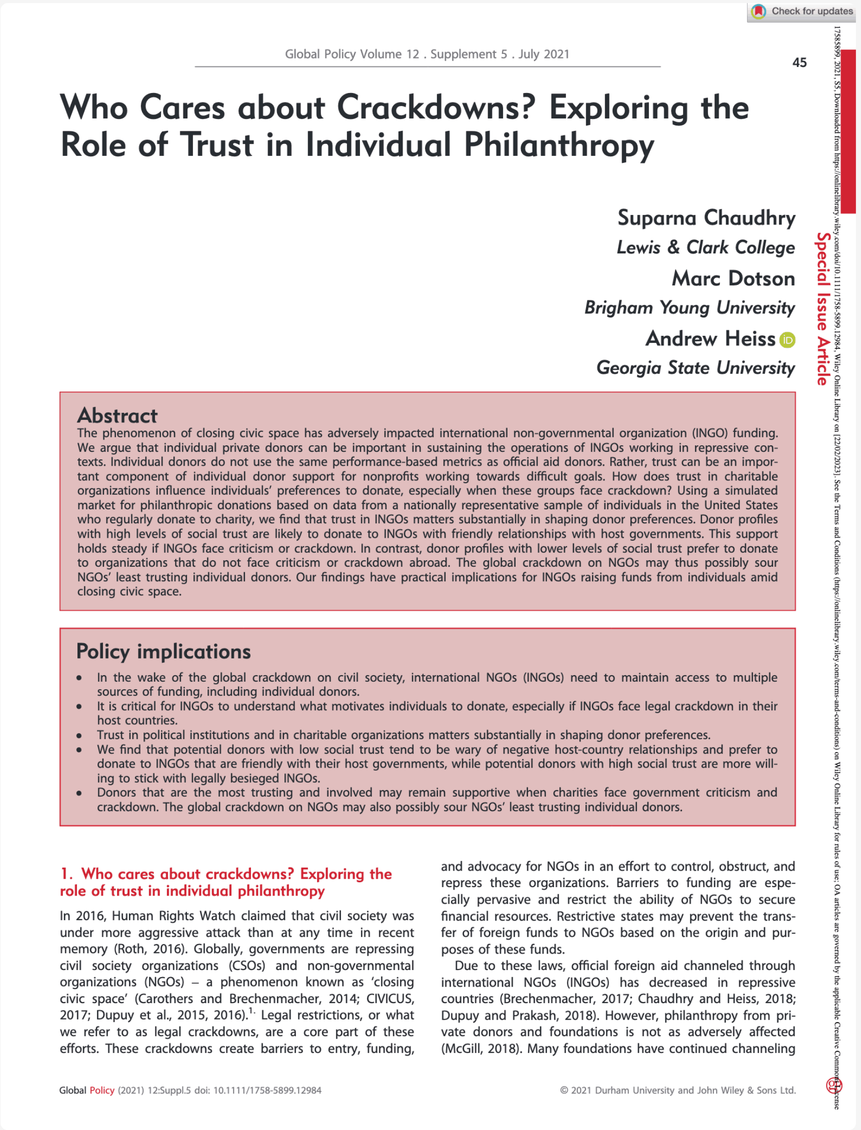
Causal inference

Social science

- Does feature X make people more or less likely to choose a “product”?
- What is the average causal effect of a specific feature?
- Estimands: **average marginal component effect (AMCE), marginal means, average feature choice probability (AFCP)**

Same data and same models, completely different estimands

Crossing traditions!



Check for updates

Who Cares about Crackdowns? Exploring the Role of Trust in Individual Philanthropy

Suparna Chaudhry
Lewis & Clark College
Marc Dotson
Brigham Young University
Andrew Heiss
Georgia State University

Special Issue Article

Abstract

The phenomenon of closing civic space has adversely impacted international non-governmental organization (INGO) funding. We argue that individual private donors can be important in sustaining the operations of INGOs working in repressive contexts. Individual donors do not use the same performance-based metrics as official aid donors. Rather, trust can be an important component of individual donor support for nonprofits working towards difficult goals. How does trust in charitable organizations influence individuals' preferences to donate, especially when these groups face crackdown? Using a simulated market for philanthropic donations based on data from a nationally representative sample of individuals in the United States who regularly donate to charity, we find that trust in INGOs matters substantially in shaping donor preferences. Donor profiles with high levels of social trust are likely to donate to INGOs with friendly relationships with host governments. This support holds steady if INGOs face criticism or crackdown. In contrast, donor profiles with lower levels of social trust prefer to donate to organizations that do not face criticism or crackdown abroad. The global crackdown on NGOs may thus possibly sour NGOs' least trusting individual donors. Our findings have practical implications for INGOs raising funds from individuals amid closing civic space.

Policy implications

- In the wake of the global crackdown on civil society, international NGOs (INGOs) need to maintain access to multiple sources of funding, including individual donors.
- It is critical for INGOs to understand what motivates individuals to donate, especially if INGOs face legal crackdown in their host countries.
- Trust in political institutions and in charitable organizations matters substantially in shaping donor preferences.
- We find that potential donors with low social trust tend to be wary of negative host-country relationships and prefer to donate to INGOs that are friendly with their host governments, while potential donors with high social trust are more willing to stick with legally besieged INGOs.
- Donors that are the most trusting and involved may remain supportive when charities face government criticism and crackdown. The global crackdown on NGOs may also possibly sour NGOs' least trusting individual donors.

1. Who cares about crackdowns? Exploring the role of trust in individual philanthropy

In 2016, Human Rights Watch claimed that civil society was under more aggressive attack than at any time in recent memory (Roth, 2016). Globally, governments are repressing civil society organizations (CSOs) and non-governmental organizations (NGOs) – a phenomenon known as 'closing civic space' (Carothers and Brechenmacher, 2014; CIVICUS, 2017; Dupuy et al., 2015, 2016).[1] Legal restrictions, or what we refer to as legal crackdowns, are a core part of these efforts. These crackdowns create barriers to entry, funding, and advocacy for NGOs in an effort to control, obstruct, and repress these organizations. Barriers to funding are especially pervasive and restrict the ability of NGOs to secure financial resources. Restrictive states may prevent the transfer of foreign funds to NGOs based on the origin and purposes of these funds.

Due to these laws, official foreign aid channeled through international NGOs (INGOs) has decreased in repressive countries (Brechenmacher, 2017; Chaudhry and Heiss, 2018; Dupuy and Prakash, 2018). However, philanthropy from private donors and foundations is not as adversely affected (McGill, 2018). Many foundations have continued channeling

Check for updates

Article

Nonprofit and Voluntary Sector Quarterly
1–27
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Sage

Navigating Hostility: The Effect of Nonprofit Transparency and Accountability on Donor Preferences in the Face of Shrinking Civic Space

Suparna Chaudhry[1], Marc Dotson[2], and Andrew Heiss[3]

Abstract

Governments across the world have increasingly used laws to restrict the work of nonprofits, which has led to a reduction in public or official foreign aid directed toward these groups. Many international nonprofits, in response, have turned to individual donors to offset the loss of traditional funding. What are individual donors' preferences regarding donating to legally besieged nonprofits abroad? We conducted a conjoint experiment on a nationally representative sample of likely donors in the United States and found that learning about host government criticism and legal restrictions on nonprofits decreases individuals' preferences to donate to them. However, organizational features such as financial transparency and accountability can protect against this dampening effect. Our results have important implications both for understanding private international philanthropy and how nonprofits can better frame their fundraising appeals at a time when they are facing restrictive civic spaces and hostile governments abroad.

Keywords

philanthropy, conjoint experiments, donor heuristics, repression, NGOs, civil society, nonprofits

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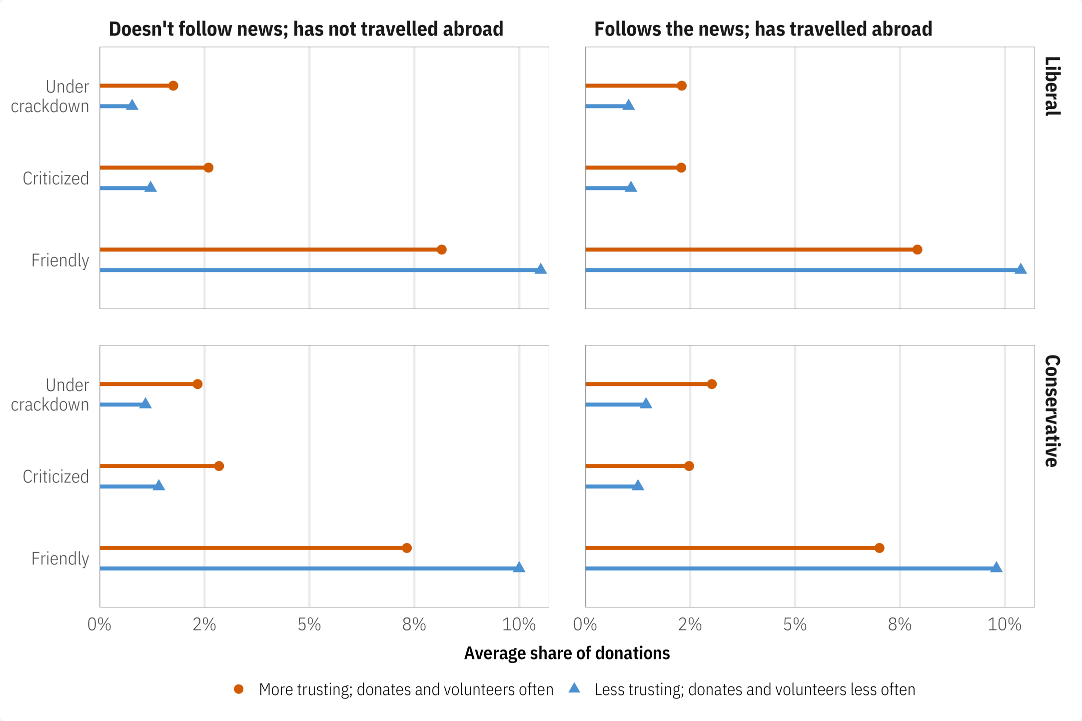
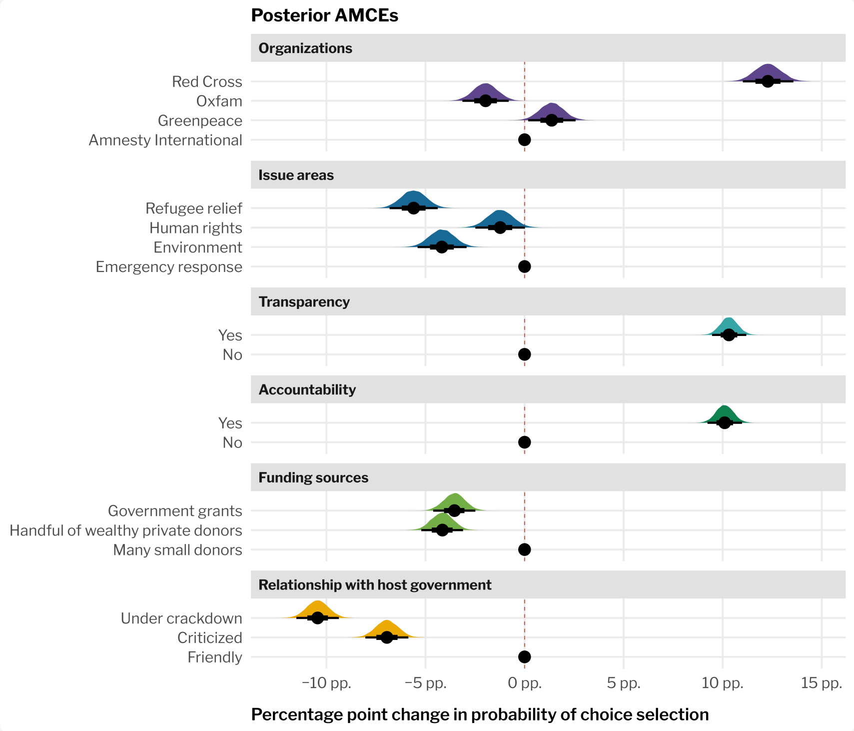
Crossing traditions!





Chaudhry, Suparna, Marc Dotson, and Andrew Heiss. 2021. “Who Cares about Crackdowns? Exploring the Role of Trust in Individual Philanthropy.” *Global Policy* 12 (S5): 45–58. https://doi.org/10.1111/1758-5899.12984.

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Roadmap for the workshop

Live coding

Hands-on practice with minimal lectures and slides per session.

We’ll work through live code together.

andhs.co/conjoint

Course outline

- ✅ ~~Intro to conjoints~~
- Designing conjoint experiments
- Conjoint data and causal inference, part 1
- Conjoint data and causal inference, part 2
- Conjoint data and respondent preferences, part 1
- Conjoint data and respondent preferences, part 2
- Implementing and fielding conjoint surveys, part 1
- Implementing and fielding conjoint surveys, part 2